

TOH YUHENG



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EDUCATION

SINGAPORE INSTITUTE OF TECHNOLOGY

Computer Science in Real-Time Interactive Simulation

2023 - Current

SINGAPORE POLYTECHNIC

Diploma in Game Design and Development

2018 - 2021

LANGUAGES

C++, C

C#

HTML, CSS, JavaScript

React, Vite

Python

English, Chinese

PROFILE

Motivated computer science student with hands-on experience developing game engines and real-time applications. Seeking an internship opportunity to further develop technical skills and contribute to meaningful projects.

WORK EXPERIENCE

INTERN

Vopak | Mar 2020 - Aug 2020

Developed an Augmented Reality (AR) application designed to assist workers in on-site learning and training.

- Designed and implemented a system that allowed workers and visitors to scan for verification, with data automatically sent and formatted in Excel for easier monitoring and tracking.
- Improved workflow efficiency by automating data collection processes and minimizing manual input errors.

PROJECTS

3D GAME ENGINE (CURRENT)

Developing a 3D game engine with a focus on advanced physics implementation.

- Responsible for PhysX integration and exploring NVIDIA FleX for responsive particle system simulations.
- Working toward creating a user-friendly and modular architecture to streamline development for gameplay programmers on the team.

SOFTWARES

Microsoft Excel

Unity

Unreal Engine

Visual Studio

Visual Studio Code

2D CUSTOM-BUILT GAME ENGINE

Designed and developed a 2D game engine from scratch, contributing primarily to the physics system.

- Implemented rigid body, collision detection and response techniques inspired by the Real-Time Collision Detection book.
- Developed the camera system and basic lighting features to enhance scene rendering and gameplay visibility.

3D PROCEDURAL CAVE GENERATION

Implemented Binary Space Partitioning (BSP) and Cellular Automata algorithms to procedurally generate realistic cave structures.

- Focused on optimizing generation performance and creating natural, varied cave formations through a set of rules for Cellular Automata.
- Integrated with Perlin Worms done by another member.

3D MULTIPLAYER GAME

Developed a 3D multiplayer "last-man-standing" party game as my final-year project in polytechnic, using Unity.

- Handled all Photon Network-related features, including player connectivity, synchronization, matchmaking, and in-game communication.
- Ensured stable multiplayer performance by managing network events, optimizing data transfer, and testing gameplay across multiple clients.
- Collaborated with the team to integrate networked gameplay with core systems such as movement, UI, particle effects, and overall game flow.

BILL SPLITTER WEB APPLICATION

Developing a React web-based bill splitting application that simplifies expense sharing using both manual input and automated receipt scanning.

- Implemented manual bill entry with dynamic participant and item-based splitting logic.
- Integrated OCR-based receipt scanning using Tesseract.js to automatically extract item names and prices from images.
- Focused on improving accuracy of text recognition and handling edge cases such as uneven splits and missing data.
- Designed with a clean, intuitive UI to ensure ease of use across devices.